

Enforcing the minimum drinking age: state, local and agency characteristics associated with compliance checks and Cops in Shops programs

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ABSTRACT

Aims To identify state, local and organizational characteristics associated with local law enforcement agencies' implementation of two dramatically different approaches to enforcement of underage drinking laws: compliance checks and Cops in Shops programs. Compliance checks use underage decoys to attempt to purchase alcohol from retail merchants, while Cops in Shops programs deploy undercover law enforcement officers in alcohol outlets to detect and cite persons under the age of 21 who attempt to purchase alcohol. **Design** Cross-sectional telephone interview conducted as part of the Tobacco Enforcement Study (TES), which examined enforcement of laws related to youth access to tobacco. **Setting** Data were collected in 1999 among law enforcement agencies in all 50 states of the United States. **Participants** Representatives of city police departments, departments of public safety, sheriffs or county police were included ($n = 920$ local agencies). **Measurements** Alcohol compliance checks and Cops in Shops programs were the primary outcomes. Covariates included state level policies (e.g. beer tax), agency resources (e.g. number of sworn officers) and community demographics (e.g. college dormitory population). **Findings** Local enforcement agencies were more likely to perform alcohol compliance checks than to have a Cops in Shops program (73.9% compared to 41.1% in cities $> 25\,000$ and 55.7% compared to 23.9% in cities $\leq 25\,000$). Conducting compliance checks for tobacco age-of-sale laws was positively associated with alcohol compliance checks and Cops in Shops (OR 3.30, $P < 0.001$; OR 1.84, $P = 0.001$, respectively). Having a Drug Abuse Resistance Education (DARE) officer was negatively related to conducting compliance checks (OR 0.67, $P = 0.03$). Special community policing units were associated with departments having Cops in Shops programs (OR 1.80, $P = 0.006$). **Conclusions** This study used a nationally representative sample of communities to better understand state and local factors that shape local law enforcement agencies' use of two distinct approaches to underage drinking enforcement. The strong link observed between tobacco and alcohol compliance checks may indicate a culture within some law enforcement agencies supporting strict enforcement of age-of-sale laws.

Keywords Age-of-sale laws, alcohol enforcement, alcohol policy, underage drinking.

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INTRODUCTION

Within a few years of the passage of the National Minimum Drinking Age Act in 1984, the minimum drinking age was standardized at 21 in all 50 United States and the District of Columbia. Although the age-21 laws are credited with moderating youth drinking and alcohol-related

crashes, there have been low levels of enforcement (Wagenaar & Wolfson 1994). Relatively low levels of funding and resources have been allocated to the promotion and enforcement of policies aimed at preventing establishments and individuals from providing alcohol to youth. Efforts aimed at apprehending underage drinkers themselves have generally received somewhat more

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attention, but enforcement efforts remain fairly inadequate, considering the frequency of underage consumption compared to the number of citations or arrests made in each year (Wagenaar & Wolfson 1994). Self-reported drinking rates among those under age 21 remain very high and contribute to higher levels of health problems associated with alcohol, including alcohol-related motor vehicle crashes, suicide, homicide and unintended sexual encounters (Johnston *et al.* 2002; NIAAA 2002). As a result, some policy makers have prioritized policies that encourage both underage people and alcohol providers to comply with the age-21 laws.

One method of enforcement of alcohol and tobacco age-of-sale laws is law enforcement agencies' use of underage decoys who attempt to purchase alcohol or tobacco from retail merchants. A number of studies suggest that these age-of-sale compliance checks, often coupled with media support, can significantly reduce the number of outlets that sell alcohol to underage people (Preusser, Williams & Weinstein 1994; Lewis *et al.* 1996; Grube 1997; Wagenaar *et al.* 2000). They have also demonstrably reduced the reported ease of acquisition by minors (Grube 1997; Wagenaar *et al.* 2000), and have resulted in significant reductions in alcohol-related health problems (Preusser *et al.* 1994). Similar findings have been established from compliance check programs using underage decoys aimed at minimum age-of-sale laws for tobacco (Forster & Wolfson 1998).

Cops in Shops, a program which involves deploying undercover law enforcement officers in alcohol outlets to detect and cite youth who attempt to purchase alcohol, is another program designed to curtail underage possession of alcohol. It was developed by the Century Council (an alcohol industry trade organization) and is being implemented in many communities across the country with the support of the National Highway Traffic Safety Administration and other funding sources, including the Office of Juvenile Justice and Delinquency Prevention's Enforcing Underage Drinking Laws Program (Century Council 2003; Wolfson *et al.* 2004). Efforts to publicize the program are implemented in conjunction with the enforcement efforts, with the intent of deterring underage alcohol purchase attempts by increasing perceptions of risk of citation and/or arrest if underage people attempt to purchase alcoholic beverages.

Despite the growing popularity of this program, we are unaware of any rigorous evaluation of Cops in Shops or point-of-purchase programs operating under different names. One study, commissioned by the Washington State Alcoholic Beverage Control Agency, uses a pre-post, no control group design, which limits inference about the success of Cops in Shops in limiting youth access to alcohol (Stern 1997). Moreover, Cops in Shops represents an approach that is fundamentally different from that of

compliance checks. In this approach, the aim is to deter youth from attempting to buy alcohol, as opposed to compliance checks' focus on deterring outlets from selling alcoholic beverages to youth.

Previous research has focused primarily on social, community and policy factors that affect the access of underage youth to alcohol with little research on state and community characteristics associated with enforcement of age-of-sale laws. In an analysis of factors that impede enforcement of the age-of-sale laws for tobacco sales in Massachusetts, DiFranza & Rigotti (1999) identified the lack of political support, budgetary constraints and organizational problems as the primary causes of reduced enforcement efforts. In a series of interviews with law enforcement officers, Wolfson and colleagues determined that officers' perceived lack of community support for alcohol age-of-sale enforcement, as well as organizational and resource constraints, made officers less likely to actively enforce alcohol age-of-sale laws (Wolfson, Wagenaar & Hornseth 1995).

Wagenaar & Wolfson (1994) examined a variety of state characteristics in order to identify predictors of higher liquor law arrest rates at the state level. They determined that the number of police officers and the rate of minor (Part 2) crimes were related inversely to alcohol law enforcement. Furthermore, they found that the rate of underage drinking within a state seemed to have no relationship with the arrest rate. They also identified several policy constraints (e.g. lack of resources for state enforcement agencies) that limit alcohol age-of-sale enforcement in the states (Wagenaar & Wolfson 1994).

In a further analysis, 295 counties in Kentucky, Michigan, Montana and Oregon were examined to identify predictors of alcohol law arrests and state Alcohol Beverage Commission (ABC) actions (Wagenaar & Wolfson 1995). The authors showed that counties with higher crime arrest rates, smaller populations, larger land areas and higher proportions of unoccupied housing (indicating the economic decline of the community) had higher rates of drinking age enforcement (Wagenaar & Wolfson 1995).

These studies form a solid base for research into the community characteristics that are associated with the implementation of two major strategies in drinking age enforcement—compliance checks using underage decoys and Cops in Shops programs. The research on community and state predictors for strict enforcement has identified three broad categories that may affect the level of enforcement: community norms and policies; agency resources and practices; and community demographics.

Community norms and policies

One of the major obstacles to successful enforcement is a perception on the part of law enforcement officers and

officials that community members do not support strict enforcement of the age-of-sale laws. Parents and community leaders may consider alcohol violations a 'lesser' crime that is not as high a priority as preventing illicit drug use and other crimes (Wolfson *et al.* 1995). In some cases many youths report that alcohol is readily available from their parents (Wagenaar *et al.* 1993). In some instances, lack of community support can lead to political pressure to ease enforcement (DiFranza & Rigotti 1999).

Community support is also crucial because implementation of some age-of-sale enforcement programs requires cooperation from alcohol merchants. Merchant cooperation is especially important for the implementation of Cops in Shops programs, where officers position themselves in the store posed as a clerk. Formal training programs for staff selling or serving alcohol, which has also been shown to reduce sales to underage drinkers, also requires the cooperation of alcohol merchants (Wolfson *et al.* 1996; Saltz & Stanghetta 1997). Overt opposition from merchants can significantly diminish uptake of these programs at the local level (DiFranza & Rigotti 1999). In circumstances where there is a perceived lack of community support, agencies may feel that stricter enforcement of the age-of-sale laws is politically dangerous and overly difficult to sustain (Wolfson *et al.* 1995).

One indicator of community attitudes is the apparent consumption of alcohol. Consumption is associated with the importance of alcohol sales to merchants and the number of outlets in a community. Apparent consumption can serve as an imperfect indicator of community member's drinking patterns. Without controlling for frequency of border crossing and the size of the tourism industry, however, such a measure cannot completely account for drinking attitudes and behaviors of residents.

Further gauges of community support for alcohol enforcement is the presence of legal obstacles to youthful drinking. While the 21-year old drinking age is now the policy of all 50 United States (Toomey *et al.* 1996), other non-universal indicators of norms include a high level of taxation on alcohol, the use of control systems of alcohol distribution and clear and stringent laws against possession. The use of a control system of distribution and high levels of taxation has demonstrably decreased the level of underage drinking (Grossman *et al.* 1994; Chaloupka & Wechsler 1996). Furthermore, unclear statutory language about possession by minors, deception of age, the use of underage decoys and consumption by minors can seriously inhibit enforcement and encourage higher rates of underage drinking (Wolfson *et al.* 1995; Toomey *et al.* 1996).

Agency resources and practices

The technical support and resources available to local law enforcement agencies can affect the implementation of

age-of-sale enforcement programs. Minimum drinking age laws may be enforced by state agencies, often the state Alcoholic Beverage Control (ABC) agency; but the onus of age-of-sale laws usually rests on local law enforcement. ABC agencies typically have limited enforcement staff (Wagenaar & Wolfson 1994) and numerous other liquor laws to enforce, and often lack jurisdiction to arrest underage people (Wagenaar & Wolfson 1995).

Limitations on local agency resources, therefore, can significantly decrease the capacity of local law enforcement agencies to implement compliance check programs. One limitation cited by officers is personnel shortages. As there is a limited number of officers on duty at any one time, enforcement of alcohol laws may be neglected due to calls that are considered higher priority (Wolfson *et al.* 1995). Previous research by Wagenaar & Wolfson (1995), however, does not fully support this argument, as the rate of officers within a population has not demonstrated a positive relationship with enforcement. This may be because agency resources may already be committed to school-based programs to prevent underage drinking.

Outside funding or support can lead to significant increases in enforcement. Implementation of compliance check programs and Cops in Shops may require that officers be assigned in either full-time or part-time capacities to enforce age-of-sale laws (Lewis *et al.* 1996; Grube 1997). Federal, state or non-government grants aimed at enforcement of drinking laws can provide this support. For example, the Robert Wood Johnson Foundation's Reducing Underage Drinking Through Coalitions program (American Medical Association 2002), as well as the Office of Juvenile Justice and Delinquency Prevention's Enforcing Underage Drinking Laws Program, can provide resources to support these and other underage drinking enforcement efforts. Finally, larger communities and agencies are more likely to have readily available resources to implement these programs.

The technical resources of the agency may also affect the likelihood of implementing these kinds of programs. Tobacco compliance checks use the same methodology, and can encounter the same political resistance, as alcohol compliance checks (Lewis *et al.* 1996). Additionally, due to US federal regulations (the Synar amendment), state that substance abuse agencies encourage local agencies to perform random, unannounced compliance checks on machines and vendors that dispense tobacco products (Forster & Wolfson 1998). It is therefore reasonable to expect that cities with programs to enforce minimum age-of-sale laws for tobacco will also find it easier to implement programs for alcohol products.

Demographics

The size of the at-risk population in a community may also affect the level of enforcement of the drinking age.

Large numbers or higher than normal rates of drinking age violators in a community may raise community awareness of the risks of underage drinkers, thus increasing enforcement efforts. High-density populations of underage youth who engage in unsafe or illegal drinking practices (e.g. college populations) may indicate large numbers of violators. (Wechsler *et al.* 2002).

Studies have also shown that white youths demonstrate higher levels of underage drinking than minority youths (Grunbaum *et al.* 2002; Johnston *et al.* 2002). In one study, limited to the greater Washington, DC area, white parents were also shown to be more permissive and the least likely to desire stricter enforcement of alcohol age-of-sale laws (Beck *et al.* 1995). Thus, the racial composition of a community could affect enforcement patterns by altering the number of liquor law violators. While such studies suggest that demographic factors could indicate patterns of enforcement within communities, we propose to test their effects in the context of more proximal determinants (e.g. agency resources) of law enforcement agencies' implementation of underage decoy programs (compliance checks) and Cops in Shops programs to reduce underage drinking.

We use a nationally representative sample of communities to assess the relationships among state policies, community characteristics and law enforcement agency characteristics, and implementation of underage decoy programs (compliance checks) and Cops in Shops programs by local law enforcement agencies. This study fills a large gap in the literature by using a nationally representative sample to examine contextual and organizational factors associated with implementation of two pre-eminent strategies for the enforcement of underage drinking and age-of-sale laws.

METHODS

Study design

This study is part of a larger project called the Tobacco Enforcement Study (TES), which examined enforcement of laws related to youth access to tobacco. Data collection methods for this study were based on previous single-state studies of enforcement of laws related to youth access to tobacco (Forster, Komro & Wolfson 1996; Cismoski *et al.* 1997). The study is a repeated cross-section in 1997 and 1999 of a nationally representative sample of communities including all 50 United States. Data include local ordinances related to youth access to alcohol and tobacco and enforcement efforts by local agencies. This study utilizes data obtained from interviewing local law enforcement agencies in 1999 and focuses on the use of compliance checks and presence of Cops in Shops programs to enforce age-of-sale laws for alcohol.

Sample

The sampling frame for this study incorporated local government units (usually cities) or units that are defined by the US Census Bureau as 'Minor Civil Divisions'. Examples of the latter include towns and townships in the New England states, New York, New Jersey and Pennsylvania, and a subset of towns and townships in Michigan, Minnesota and Wisconsin. Additionally, communities needed to have full municipal powers and a population of 2000 or greater to qualify for selection into the study.

The sample consists of 1007 communities selected from 9300 eligible communities (10.75%). The goal was to have approximately equal variance in each state so as to yield equally precise state estimates of enforcement rates, which led to sampling a minimum of 10 communities in small states and a maximum of 26 communities in large states. Because we expected enforcement patterns to vary by community size, we stratified the sample within state by community size (less than 25 000 versus 25 000 or more. Using earlier (1993) enforcement data on Minnesota communities (Forster *et al.* 1996), we used Monte Carlo methods to determine the optimal within-state stratified sampling scheme. It was determined that a compromise between sampling a fixed number of communities within population size strata and using the natural proportion of communities in each stratum as a sampling fraction yielded the most statistically desirable properties (minimal bias and reduced variance).

Telephone interviews were attempted with 1178 agencies, including the police department in each of 1007 local communities, as well as other agencies that the city clerk reported to be involved in enforcement of laws related to youth access to tobacco. A total of 1046 responded (89%). Respondents included representatives of city police departments or departments of public safety (the police chief or a designee) ($n = 878$), sheriffs or county police ($n = 99$) and other local officials ($n = 69$), such as heads of licensing agencies and local health department staff. This survey was designed to elicit information regarding enforcement efforts, enforcement strategies, frequency of enforcement efforts, number and type of actions taken against establishments or individuals found to be non-compliant and perceived barriers to enforcement of tobacco-related laws.

For analytical purposes we excluded the following: surveys completed by agencies other than the municipal police departments (e.g. health departments) ($n = 69$), units with no sworn officers ($n = 4$) and communities located in 'dry' counties ($n = 35$). Additionally, cases were excluded from the alcohol compliance checks analysis if respondents stated they did not know if their agency used underage decoys to perform compliance

checks ($n = 15$) or refused to answer ($n = 3$). A total of 920 cases were available for the alcohol compliance checks analyses and 938 cases were available for the Cops in Shops analyses.

Prevalence data on alcohol compliance checks and Cops in Shops programs were obtained to measure different strategies for enforcement activities to deter youth access to alcohol. The law enforcement agents were asked if there had been any compliance checks using an underage decoy to enforce the laws against alcohol sales to people under the age of 21 in the past 12 months (1 = yes, 0 = no). They were then asked if their department participated in Cops in Shops, or similar programs, in which undercover officers were stationed in businesses to detect alcohol purchase attempts by people under the age of 21 in the past 12 months (1 = yes, 0 = no).

Agency resources and practices

Respondents to the enforcement survey were asked if they had an officer participating in the Drug Abuse Resistance Education (DARE) program or if they had a special unit with officers assigned to neighborhood or community crime prevention, such as Neighborhood Watch or Operation ID (1 = yes, 0 = no). Respondents who did not know if they had these programs or who refused to answer were assumed to not have them in place ($n = 2$). Respondents were also asked to indicate the number of sworn personnel (i.e. officers working in the field) in their agency. In five cases where the respondent indicated that there were over 996 (due to the way the response options were categorized), data were later replaced with the actual number of sworn officers using data compiled by the Bureau of Justice Statistics (Reaves & Goldberg 1999).

Additionally, we examined whether local law enforcement agents who performed tobacco compliance checks using underage decoys were more likely to engage in alcohol enforcement efforts (1 = yes, 0 = no). We evaluated the models by coding responses of don't know ($n = 11$) or refused ($n = 5$) as a negative response or as missing and compared the models. As there was no noticeable difference in the models, these responses were coded as negative responses to maximize statistical power.

State level norms and policies

Other state norms and policies that may influence local law enforcement activities surrounding youth access to alcohol were also included in the analysis. These policies include the taxation level on alcohol and the distribution system in each state (Wagenaar *et al.* 2000). A dichotomous variable at the state level was created for the distribution system of each state obtained from *Alcohol Policies in the United States* (Alcohol Epidemiology Program 2000). States with control systems—where the off-

premises sale of liquor (and in some cases wine and strong beer) are restricted to state run outlets—were coded as 1, with states with license systems coded as 0.

The rate of taxation on alcoholic beverages was also collected from alcohol policies. This tax rate was normalized at number of cents per drink, measured as a 12-ounce serving for beer, a 1.5-ounce serving for spirits and a 5-ounce serving for wine across all 50 states (Wagenaar *et al.* 2000). However, states with controlled distribution do not have set levels of taxation on liquor, and in some cases wine, because the prices are set by the state ABC agency. Thus, only the tax on beer and other malt beverages was used in the analysis.

The 1997 per-capita consumption of alcohol was obtained from the Beer Institute's online *Brewers Almanac* (Beer Institute 2001). Consumption was calculated by dividing the apparent consumption, as measured by tax collection, by the total state population (obtained from the 1990 Census). Because apparent consumption was highly correlated across categories (malt beverages, distilled spirits and wine), we selected the per-capita consumption of malt beverages for use in the models.

We also evaluated the influence of external grant funding on local law enforcement efforts surrounding alcohol use. The Robert Wood Johnson Foundation provides grants to statewide coalitions to work against underage drinking (American Medical Association 2002). These coalitions have successfully lobbied their state governments to enact laws requiring keg registration and moving alcohol billboard advertisements further from schools. They have also run advertisement awareness campaigns and have provided knowledge and resources state-wide to groups that want to reduce underage drinking. A dichotomous variable was created where all communities in states receiving these grants were coded as 1.

Community-level demographics

Local crime statistics were compiled into Part 2 and Part 1 crimes as defined in the Bureau of Justice Statistics (Pastore & Maguire 2000). Inclusion of more than one crime index in the models resulted in high levels of multicollinearity. Wagenaar & Wolfson (1995), in their study of liquor law enforcement levels at the county level, previously found a significant positive relationship between arrest rates for minor crimes and alcohol law enforcement. Building on that finding, Part 2 crimes (forgery, fraud, vandalism, drug abuse, etc.) were used in the final models. Additional demographic characteristics of the local communities were included in the models. The 1990 census data were used for the demographic characteristics of the community, including the median family income in thousands of dollars, the percentage of the population that was non-white and the log of the total

population, all continuous measures. 1990 census data were also used to create a categorical variable measuring local college dormitory population (no college dorm population, college dorm residence of 0–1999 per 100 000 people and 2000 + college dorm residence per 100 000 people).

Analysis

Two logistic regression models were computed, with alcohol compliance checks and Cops in Shops as the dependent variables. This allowed us to compare the association of agency characteristics, state and local norms, and community demographics with two different types of alcohol enforcement efforts. Because we had difficulty obtaining complete data on Part 2 crimes, median income and college dormitory population, multivariate models were recomputed without these variables, increasing the sample size for the analyses. Stata 7.0 was used to conduct the analyses. We also conducted all analyses using the SVYLOGIT analytical procedure. This approach controls for clustering associated with sampling multiple communities within a given state (StataCorp 2001).

RESULTS

Local enforcement agencies were twice as likely to perform compliance checks as Cops in Shops programs. Moreover, agencies in larger communities were more likely to conduct these operations than those in smaller communities. Almost three-quarters (73.9%) of agencies in larger communities (> 25 000 population) reported conducting compliance checks in the 12 months preceding the survey, compared to 55.7% in smaller communities ($\leq 25\,000$). Four in 10 (41.1%) of agencies in larger communities reported conducting Cops in Shops operations, compared to 23.9% of agencies in smaller communities. About one-quarter of the agencies engaged in both efforts and about one-third (31%) reported having neither program.

Factors associated with enforcement

Table 1 presents the results from the logistic regression analyses.

The data did not show a relationship between state policies and the implementation of alcohol compliance checks. However, the level of taxation on beer was marginally related to the likelihood of having a Cops in Shops program in place [OR = 1.09 (0.99–1.20), $P = 0.09$].

Agency resources and practices were related more strongly to the likelihood of performing compliance checks and having a Cops in Shops program. Agencies that engaged in tobacco compliance checks were 3.3 times more likely to engage in alcohol compliance checks

[OR = 3.30 (2.39–4.56)] and almost two times more likely to have a Cops in Shops program [OR 1.80 (1.18–2.73)]. Agency participation in the DARE program showed a negative relationship to alcohol compliance checks [OR = 0.67 (0.47–0.96)], yet was marginally (and positively) related to having a Cops in Shops program [OR = 1.38 (0.95–2.00)]. The relationship between a special community policing unit approached statistical significance ($P = 0.08$) in the alcohol compliance checks model and was related to agencies participating in Cops in Shops [OR = 1.80 (1.18–2.73)].

Larger population size was also associated with compliance checks [OR = 1.47 (1.20–1.78)] controlling for agency characteristics (e.g. number of sworn officers) that are directly related to the size of a community. Finally, law enforcement agencies in communities with a high density of college dormitory populations were almost two times more likely to have a Cops in Shops program in place [OR 1.85, (1.13–3.02)]. College population was unrelated to alcohol compliance checks.

DISCUSSION

Alcohol enforcement efforts appear to be associated with resources available to local law enforcement agencies. Conducting compliance checks for tobacco age-of-sale is strongly and significantly related to the agencies' propensity to perform similar checks on alcohol sales. Furthermore, performing compliance checks was negatively related to having a DARE officer. This finding is consistent with the notion that the presence of alternative substance use prevention programs detracts from resources devoted to more effective methods of prevention (Ennett *et al.* 1994). Having a special unit in the department for community policing seems to be positively related to implementing these programs, as expected.

In addition, we found that more populous communities were more likely to have underage decoy programs. This may reflect the greater availability of resources in larger communities allowing for more compliance checks. Because we also controlled for the rate of sworn officers within the community, a larger population also indicates a larger enforcement agency which may have greater differentiation of functions, such as set programs for age-of-sale enforcement.

Law enforcement agency resources and practices were also related to Cops in Shops. The presence of tobacco age-of-sale compliance checks was significantly related to this enforcement strategy, as was having a special policing unit. Societal norms were also related to the Cops in Shops program through the level of beer taxation. The positive relationship of the level of beer taxation to the Cops in Shops program is consistent with the argument that the beer tax is an indicator of the

Table 1 Logistic regression analyses of state, local and agency characteristics that influence alcohol compliance checks and Cops in Shops programs in local law enforcement agencies ($n = 790$).

	$(n = 790)^1$			
	Alcohol compliance checks		Cops in Shops	
	OR, 95% CI	P	OR, 95% CI	P
State level norms and policies				
Per capita consumption of malt beverages in gallons	0.99 (0.95–1.04)	0.29	1.00 (0.96–1.05)	0.96
States with controlled distribution systems	0.88 (0.63–1.24)	0.67	0.77 (0.54–1.09)	0.14
Community in a state that received a RWJF reducing underage drinking through coalitions grant	1.24 (0.83–1.85)	0.29	1.03 (0.56–1.92)	0.41
Cents of tax on one 12-ounce drink of beer	1.07 (0.96–1.19)	0.21	1.09 (0.99–1.20)	0.09
Agency resources and practices				
Sworn officers per 100 000 people	1.00 (1.00–1.00)	0.09	1.00 (1.00–1.00)	0.81
Agencies that conduct tobacco sale compliance checks using underage decoys	3.30 (2.39–4.56)	< 0.001	1.84 (1.30–2.60)	0.001
Agencies with a DARE officer	0.67 (0.47–0.96)	0.03	1.38 (0.95–2.00)	0.09
Agencies with a special community policing unit	1.39 (0.96–2.01)	0.08	1.80 (1.18–2.73)	0.006
Community level demographic variables				
No college dorm population (%)	1.00		1.00	
College dorm population between 1 and 1999/100 000 people (%) (referent)	0.68 (0.38–1.22)	0.20	1.40 (0.83–2.36)	0.21
Dorm population greater than 2000 per 100 000 people (%)	1.00 (0.59–1.71)	0.97	1.85 (1.13–3.02)	0.01
Log of the population of the community	1.47 (1.20–1.78)	< 0.001	1.11 (0.92–1.34)	0.29
Percentage of the population that is non-white	1.00 (0.99–1.01)	0.44	1.00 (0.99–1.01)	0.86
The median household income of the community in thousands of dollars	0.98 (0.97–1.00)	0.01	1.00 (0.98–1.01)	0.72
Part 2 crimes per 100 000 people	1.00 (1.00–1.00)	0.38	1.00 (1.00–1.00)	0.96

¹Due to missing data on crime, college dorm population, and income, the data available for multivariate analyses were reduced. Data were reanalyzed without Part 2 crime data, income, and college dorm population ($n = 911$), but the results did not change meaningfully, with one exception. Law enforcement officers with a special policing unit were more likely to perform alcohol compliance checks.

latent variable measuring community support for strict enforcement. The use of Cops in Shops programs is also related to the presence of a large visible at-risk population, in the form of college students, while the compliance checks model has no significant relationship with these variables. This may indicate that more visible populations of drinking age violators may lead to more violations and calls for stricter enforcement aimed at youthful offenders rather than targeting alcohol vendors. Alternatively, it is possible that the Century Council has specifically targeted areas with large concentrations of college students in marketing this program to retailers.

Future examinations of correlates to age-of-sale enforcement programs could improve on these models in a number of ways. Future surveys of communities should seek to more directly gauge the supportiveness of the community for age-of-sale enforcement programs as well as the presence of coalitions advocating their use. In addition, more detailed information on state and local policies towards underage possession and laws potentially influencing the use of underage decoys would be useful. A bet-

ter measure of the importance of alcohol sales to the local community may reveal underlying economic considerations that affect policy makers' decisions regarding policing underage alcohol sales. Finally, future research would benefit from more detailed data on compliance checks (e.g. are they focused on premise or off-sale outlets) and on additional enforcement strategies (e.g. 'shoulder tap' operations).

The finding that only 55.7% of law enforcement agencies in smaller communities ($\leq 25\ 000$ population) and 73.9% of agencies in larger communities ($> 25\ 000$ population) had conducted even a single compliance check in the year preceding the survey is cause for concern. Compliance checks are widely acknowledged to be the linchpin of efforts to promote merchant compliance with alcohol age-of-sale laws. It remains to be seen whether efforts such as the US Office of Juvenile Justice and Delinquency Prevention's 50-state, Enforcing Underage Drinking Laws Program results in increases in the use of compliance checks by local law enforcement agencies (see Wolfson *et al.* 2004).

We also found that four in 10 agencies in larger communities (41.1%) and about one-quarter of agencies in smaller communities (23.9%) had conducted Cops in Shops programs in the year preceding the survey. Given the widespread diffusion of this program, systematic evaluation of its effectiveness in reducing underage drinking is urgently needed. Moreover, the fact that Cops in Shops programs are relatively rare in communities that are not also implementing compliance checks suggests that they may arise in response to compliance check programs. The concern is that this unproven approach may compete for limited resources with compliance check programs.

This study is the first to utilize a nationally representative sample of communities to better understand state and local factors that affect enforcement of underage alcohol sales through two pre-eminent strategies: compliance checks and Cops in Shops programs. The strong link observed between tobacco and alcohol compliance checks may indicate a general culture or philosophical approach within a law enforcement agency regarding strict enforcement of age of sale laws and other public health measures. More information is needed on the agency characteristics that affect adoption and implementation of laws that restrict underage youth from tobacco and alcohol.

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